

Skanda Bharadwaj

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Education

Pennsylvania State University

Ph.D. in Computer Science and Engineering

Aug 2021 – Present

M.Sc. in Computer Science and Engineering

Jan 2019 - May 2021

PES Institute of Technology

B.Tech in Telecommunications Engineering

Aug 2011 – May 2015

Specialization

Computer Vision, Deep Learning, Recommendation Systems, Content Understanding

Work Experience

Machine Learning Engineer Intern @ Meta – Menlo Park, CA

May - Aug 2026

- Currently working on model-based systems for content cold start optimization, including designing Transformer-based prediction models to optimize content delivery strategies and improve the creator experience.

Machine Learning Engineer Intern @ Meta – Bellevue, WA

May - Aug 2025

- Designed and implemented the A2A2U and A2U2U recommendation pipelines, leveraging KNN, LLM-based embeddings, and user/creator clustering to solve the cold start problem for emerging creators.
- Achieved >99.9% creator coverage improvement (26K+ newly onboarded creators) in offline experiments, delivering scalable outputs integrated directly into Meta's recommendation systems.

Research Scientist Intern @ SLB-Doll Research – Cambridge, MA

May – Aug 2024

- Developed a real-time image processing pipeline that integrates traditional techniques with deep learning classifiers to segment, detect, and localize methane plumes for enhanced environmental monitoring.

Computer Vision Engineer @ Continental – Bengaluru, India

July 2015 – Dec 2018

- Played a key role in the development of multi-object tracking For Advanced Driver Assistance Systems (ADAS).

Research

Research Assistant @ [LPAC](#)

Aug 2021 – Present

- **Learning from human perception for visual pattern understanding:** Leveraging human-annotated recurring patterns to learn perceptually grounded visual representations for scene understanding. The research connects human visual grouping behavior with machine learning models for recurring pattern discovery and structured image understanding.
- **Computer vision for geometry and human stability analysis:** Developing vision-based models for vanishing point detection and cross-modal human stability estimation, including methods that connect image structure, pose, and foot-pressure dynamics for interpretable visual reasoning.

Research Assistant @ [UIT Lab](#)

Jan 2019 – May 2021

- **Ultrasound Image Tracking Enhancement:** Investigation into the integration of Siamese convolutional neural networks with Kalman filtering to enhance tracking accuracy in ultrasound imaging.

Selected Publications [Google Scholar](#)

- **Skanda Bharadwaj**, R. T. Collins and Y. Liu, "Recurrence-based Vanishing Point Detection", In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* February 2025, pp. 8909–8918. [[Paper](#)]
- K. Y. Kraiger, J. Li, **Skanda Bharadwaj**, Y. Liu, and R. T. Collins, "Estimating Foot Pressure and Stability from Visual Input", In *36th British Machine Vision Conference 2025, BMVC*, Sheffield, UK, November 24-27, 2025. [[Paper](#)]
- Shimian Zhang, **Skanda Bharadwaj**, Keaton Kraiger, Yashasvi Asthana, Hong Zhang, Robert Collins, and Yanxi Liu, "Novel 3D Scene Understanding Applications From Recurrence in a Single Image", *arXiv preprint* 2022. [[Paper](#)]
- **Skanda Bharadwaj**, Sumukha Prasad, Mohamed Almekkawy, "An Upgraded Siamese Neural Network for Motion Tracking in Ultrasound Image Sequences", *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 68(12):3515-3527, 2021. [[Paper](#)]